Mobile App Update Impact and Failure Detection

Final Project Presentation

Team 10

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Guide / Faculty: Rohini S.

Presentation date: 20/10/2026

Business problem and objectives

Business problem: Product and support teams of 11 Indian consumer apps (Swiggy, Flipkart, Paytm and others) receive 117 to 1,539 Play Store reviews a day. Failed payments, refunds, late deliveries and bad updates are noticed late, because nobody can read them all.

Objectives: (1) Flag problem reviews automatically, far above the 19.5% base rate. (2) Find what users complain about, forecast daily 1–2★ reviews per app, and test every app release for harm.

Business decision: Which app, complaint and release to fix first, and how much support capacity to plan for next week.

Expected output: A problem-review classifier, complaint topics with sentiment, 4-week forecasts with unusual-day alerts, release verdicts and a per-app priority table with 65 recommendations, all in a live dashboard.

Complete project workflow

Stage	Your implementation
Unit 1	Own Play Store collection (1,202,729 raw → 1,137,987 validated reviews, 11 apps); cleaning; 9 keyword issue tags; VADER sentiment; EDA with 15 charts and bias checks
Unit 2	TF-IDF → 200 SVD components + 15 structural features → Logistic Regression and Random Forest, tested on 227,598 held-out reviews (PR-AUC 0.87)
Unit 3	Method 1: Text Mining (NMF topics + VADER) and Method 2: Time-Series Analysis (ARIMA/SARIMA forecasts + release impact)
Final application	Streamlit dashboard, 6 pages, live at team10-app-reviews.streamlit.app; priority table and 65 recommendations

Completion status: 100% implemented and integrated: every stage merged on GitHub, 29 of 29 board tasks Completed (15 in Review 1, 14 in Review 2).

Overall dataset description

Source and collection method: Google Play Store, collected by us with the google-play-scraper package: every English review from India, newest first, for 11 apps. No Kaggle, UCI or GitHub dataset.

Final dataset size: 1,137,987 reviews × 29 variables (11 gzipped CSV files in data/tagged/; dictionary in DATA_DICTIONARY.md).

Coverage: 1 April – 20 September 2026 (173 days), same window for every app. Food & Grocery: Swiggy, Zomato, Blinkit, Domino's. Shopping: Myntra, Flipkart, Amazon, Meesho. Payments: Paytm, PhonePe, Google Pay.

Unit mapping: Unit 1: all 29 variables. Unit 2: content, 9 issue flags, 4 VADER scores, length and upvotes → target score ≤ 2. Unit 3 text mining: content of 1–2★ and 4–5★ reviews. Unit 3 time series: review_date, score and app_version → 11 daily series and 224 releases.

FINAL PROJECT DATA

Sample dataset and key variables

app_name	content (review text)	primary_issue	sentiment_compound	score → is_problematic
Amazon	very bad service, no responsibility	customer_support	−0.74	1★ → 1
Flipkart	unexcepted delivey cancellation issue	cancellation_return	0.00	1★ → 1
Zomato	Worst customer care support no customer care number	customer_support	+0.48	1★ → 1
Myntra	best shopping app and products are very good. fast dilivery	none	+0.81	5★ → 0

Key variables: score (integer 1–5); content (text); review_date (datetime); app_version (text, 13.4% empty); 9 issue flags (0/1, keyword rules); VADER compound (−1 to +1); full list of 29 in DATA_DICTIONARY.md

Method-specific subsets: text mining uses 167,357 complaint and 166,929 praise reviews with ≥ 3 content words; time series uses daily 1–2★ counts (11 apps × 173 days) and 224 release events. Sample: data/sample/final_dataset_sample.csv (110 rows)

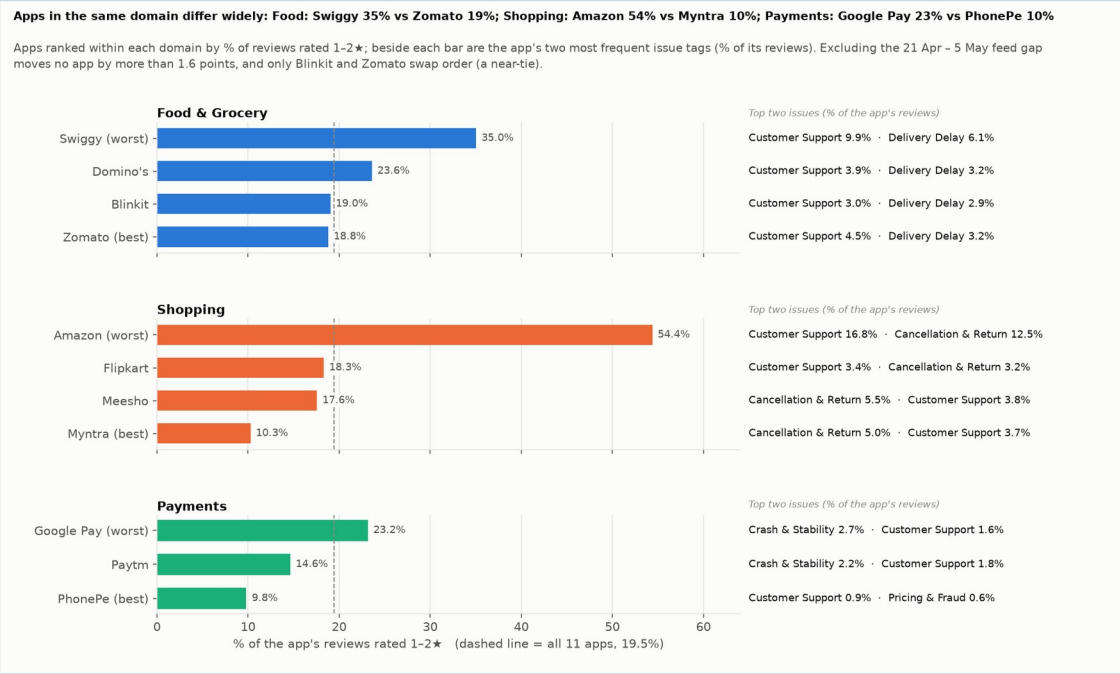
OVERALL VALIDATION FOR UNITS 1 TO 3

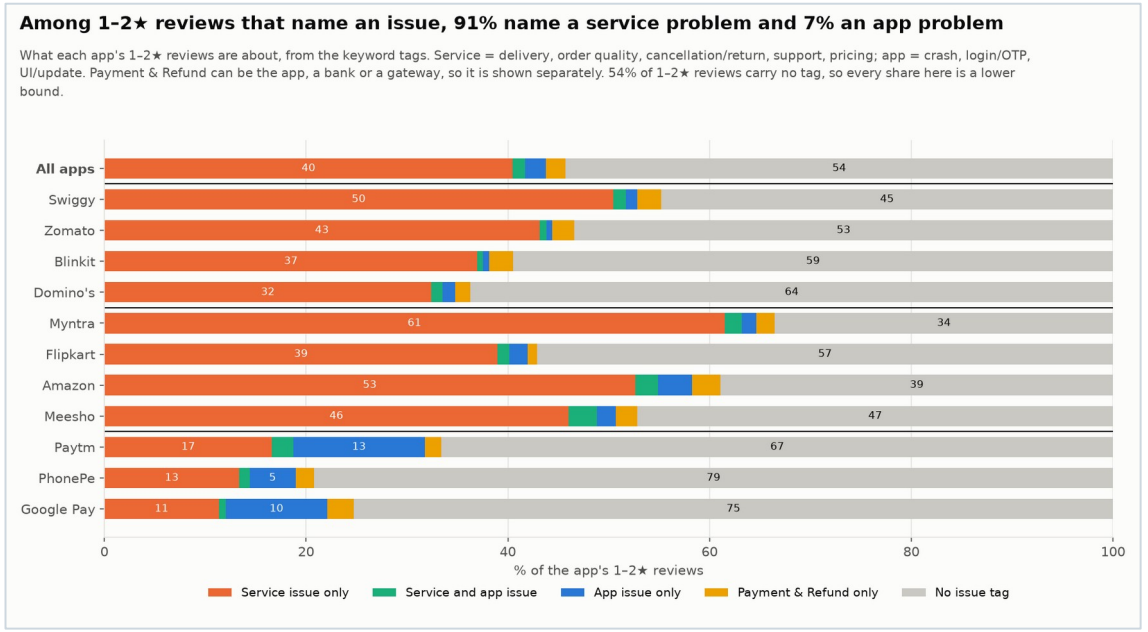
Dataset validation and preparation

Check	Finding	Correction / evidence
Missing values and duplicates	0 duplicate review_id; 47,533 empty or < 3-character reviews, 7,420 without letters, 9,789 non-English; app_version empty for 13.4%	Removed (1,202,729 → 1,137,987); app_version kept, not a model input; 0 missing in the other 28 columns
Types, ranges and consistency	Ratings 1–5, dates in window, derived fields (month, length, tag counts, sentiment labels) agree with their sources	18 automated checks, all pass: scripts/24_dataset_documentation.py → DATA_VALIDATION.md
Suitability across all units	Feed gap 21 Apr – 5 May drops positive reviews for 5 apps; Zomato has no reviews on 24 Jul	Flagged, not hidden: counts (not shares) for time series, gap as an ARIMA input, Zomato day filled for forecasting only
Synthetic data, if applicable	None: every record is a real Play Store review	Not applicable

Final validated dataset: 1,137,987 of 1,202,729 raw reviews retained (94.6%) · data/tagged/<app>.csv.gz (11 files, 29 variables)

Exploratory analysis and preprocessing





Key findings: (1) Apps differ widely even within a domain: 54% of Amazon's reviews are 1–2★ vs 10% for Myntra; Swiggy 35% vs Zomato 19%. (2) 91% of 1–2★ reviews that name an issue describe a service problem (support, delivery, refunds, returns), only 7% an app fault.

Preparation: duplicates, empty, emoji-only and non-English reviews removed; 9 keyword issue flags and VADER scores added; the late-April feed gap flagged so it is not read as a wave of complaints.

Predictive modelling and final evaluation

Model	Precision	Recall	F1	PR-AUC
Baseline: always 'not a problem'	0.000	0.000	0.000	0.195
Logistic Regression	0.770	0.858	0.811	0.871
Random Forest (selected)	**0.736**	**0.870**	**0.798**	**0.874**

Evaluation setup: target is_problematic = 1–2★ (19.5%); inputs: 200 SVD text components + 15 structural features; stratified 80/20 split, 910,389 training and 227,598 held-out test reviews. TF-IDF, SVD and the scaler are fitted on training rows only (scripts/25); this changes no metric by more than 0.003 against Review 1. Out-of-time test (train Apr–Aug, test Sep): PR-AUC 0.866.

Selected model and interpretation: Random Forest for triage: it finds 87% of problem reviews and about 3 in 4 flags are real; Logistic Regression is close with fewer false alarms. Limitation: the target is the user's rating, not a confirmed defect.

UNIT 3

Selected advanced analytics methods

Selected method	Business question	Data used
Method 1: Text Mining	What are users complaining about in each domain and app, and which complaints are the angriest?	content of 1–2★ reviews with ≥ 3 content words (167,357), 4–5★ reviews as a contrast (166,929), VADER scores, the 9 keyword tags for comparison
Method 2: Time-Series Analysis	How many 1–2★ reviews will each app get in the next weeks, which days are unusual, and did a new release make things worse?	review_date + score → daily 1–2★ counts per app (11 × 173 days); app_version → 224 release events

Selected: Time-Series Analysis and Text Mining. Association rules not chosen: only 3.4% of reviews carry two or more issue tags.

Method 1 implementation

Selected method: Text mining with NMF topic models and VADER sentiment: find the complaint themes in each domain and how negative each one is.

Input and preparation: 167,357 complaint reviews (1–2★, ≥ 3 content words). Lowercase; keep negations ("not delivered"); Hinglish "nahi" → "not"; drop app names and brand phrases; lemmatise. One document-term matrix per domain with words and two-word phrases (in ≥ 10 reviews and ≤ 50% of them).

Implementation: scikit-learn: TF-IDF weights → NMF (nndsvda start, seed 42), K = 6 / 6 / 5 topics for Food & Grocery / Shopping / Payments, chosen by NPMI coherence over K = 3–10. Log-odds with an informative prior for words distinctive to each app. Pipelines saved for the dashboard.

Output: 17 complaint topics, each with top terms, example reviews and sentiment: Order Cancellation & Refund is 35% of food complaints (VADER −0.27), Return Rejection & Defective Products 25% of shopping, Failed Transfers & Debited Amounts 52% of payments; angriest: Scam Allegations (−0.61, 80% strongly negative).

UNIT 3

Method 1 evaluation and comparison

Approach / configuration	Held-out NPMI ↑	Topic diversity ↑	Largest topic share
NMF, TF-IDF (selected) · Food & Grocery	**0.192**	**0.92**	34%
LDA, counts · Food & Grocery	0.117	0.68	23%
NMF, TF-IDF (selected) · Shopping	**0.164**	**0.88**	26%
LDA, counts · Shopping	0.120	0.73	26%
NMF, TF-IDF (selected) · Payments	**0.152**	**0.94**	50%
LDA, counts · Payments	0.076	0.76	29%

Evaluation data: the same document-term matrices and K for both models; each fitted on 80% of a domain's complaints (133,884 reviews) and scored on the unseen 20% (33,473). K itself chosen by NPMI over K = 3–10. Topics vs the 9 keyword tags: 71,929 complaints (43%) match no tag but get a topic.

Interpretation: NMF gives more coherent and more distinct topics in every domain, so it is selected; LDA spreads reviews more evenly but mixes themes. Business meaning: refunds, returns and failed transfers are the largest fixes; loss and broken trust drive the angriest reviews. Limitation: topic names are our reading of the top terms; VADER underrates calm technical complaints.

Method 2 implementation

Selected method: Time-series analysis: forecast daily 1–2★ reviews per app, flag unusual days, and test whether each new app version made things worse.

Input and preparation: daily 1–2★ counts per app, 11 series × 173 days (1 Apr – 20 Sep). Counts, not shares, because of the feed gap, which enters the model as a 0/1 input; Zomato's missing 24 Jul filled. ADF test: differencing only for Zomato and Blinkit. 224 release events from app_version.

Implementation: statsmodels SARIMAX: weekly seasonal terms (lag 7) for the 4 food apps (Sundays +22–38%), orders from ACF/PACF and lowest AICc (378 fits). Time split: rolling origin, train on the first 84 days, forecast 28, move one week (9 origins). Baselines: seasonal naive and Holt-Winters. Releases: new- vs older-version 1–2★ share over the same 7 days, Benjamini–Hochberg corrected.

Output: 4-week forecast with 80% and 95% intervals, e.g. 21–27 Sep: Flipkart 1,830 (1,596–2,056), Swiggy 1,037 (889–1,180); 83 unusual days flagged; 4 of 205 releases worse, Paytm 10.84.5 confirmed (34 → 66 1–2★ per day).

Method 2 evaluation and comparison

Approach / configuration	MAE ↓	RMSE ↓	MASE ↓	Beats baseline
ARIMA / SARIMA (selected)	**17.95**	**24.95**	**0.88**	**10 of 11 apps**
Holt-Winters (damped trend + weekly season)	19.14	26.66	0.93	–
Seasonal naive (same weekday last week)	21.99	31.07	1.08	baseline

Evaluation data: future held-out days only: 9 rolling origins × 28 days = 252 forecasts per app and model (2,772 per model), orders chosen on training days; mean over the 11 apps (1–2★ reviews per day). 95% intervals contain 93% of days; Ljung–Box finds no residual autocorrelation for 11 of 11 apps.

Interpretation: ARIMA is the most accurate and the only model with intervals, so it drives capacity planning and daily alerts. Zomato, Blinkit and Domino's are the most predictable; Google Pay is the one app where the baseline wins, and Paytm's September jump makes its forecast unreliable. Limitation: 173 days, no yearly season; level shifts cannot be predicted.

BUSINESS OUTCOMES

Combined insights and recommendations

Evidence from the project	Recommended action	Expected benefit / limitation
Unit 2 + Unit 3: the classifier finds 87% of problem reviews; topics name them, including the 43% no keyword tag matched	Route flagged reviews by topic to the team that owns it (refunds, returns, payments, support)	Faster triage; assumes 1–2★ means a real problem; Paytm flags only 66% precise, so check by hand
Both Unit 3 methods: Paytm 10.84.5 users 23% vs 13% 1–2★, app rose 34 → 66 a day, 6 forecast alerts that week	Investigate and patch Paytm 10.84.5 first; test every release against the app's usual gap	Stops the September rise; three other worse releases overlap the feed gap
Text mining: refunds 35% of food complaints, returns 25% of shopping, failed transfers 52% of payments	Fix these flows first in every app of the domain; answer scam allegations quickly (−0.61)	Largest volume reduction; topic shares cover reviews with ≥ 3 words
Six warning signs from all stages: High priority = Swiggy, Flipkart, Amazon, Paytm	Focus on these four; staff support for the forecast (Swiggy ~1,037 1–2★ next week)	Equal weights and fixed thresholds; every sign is shown

Business objective addressed: teams can now see what is going wrong, whether it is getting worse, whether an update caused it, and what to fix first: 65 recommendations, each backed by a number.

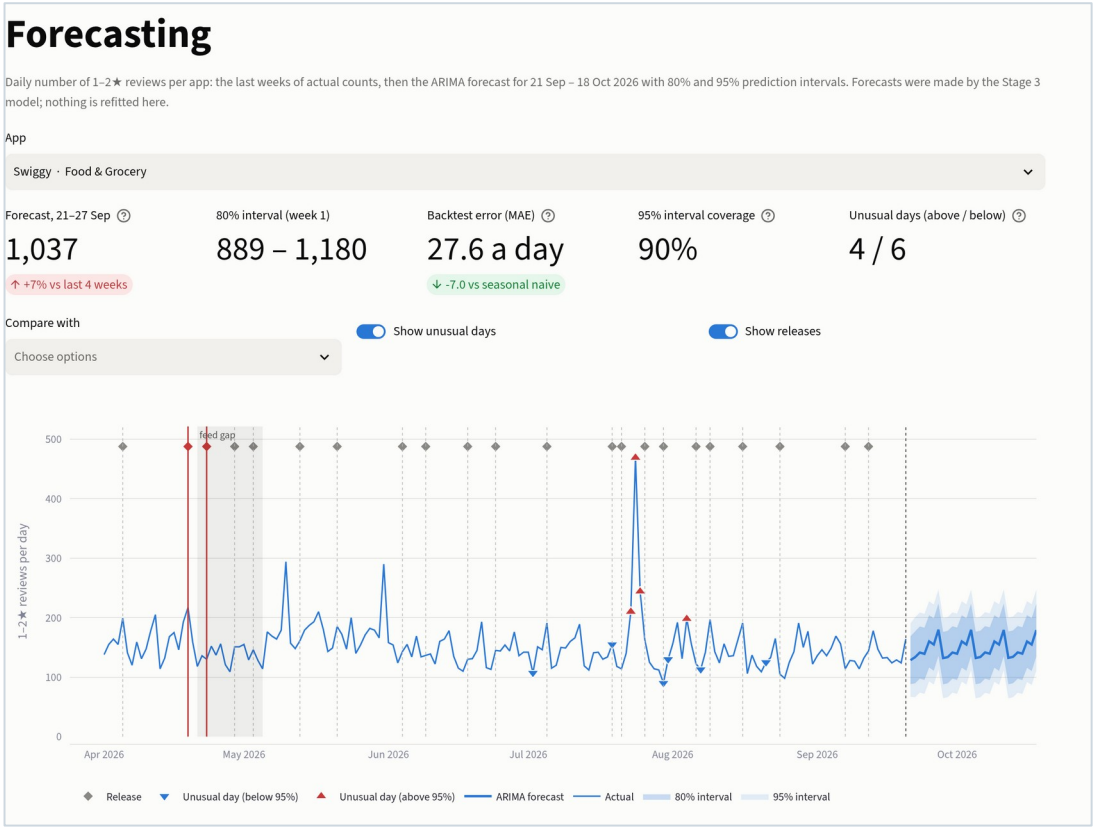
MANDATORY LIVE DEMONSTRATION

Interactive dashboard or website

Application: Streamlit dashboard, 6 pages (overview, text mining, forecasting, update impact, recommendations, check a review): team10-app-reviews.streamlit.app · local: streamlit run dashboard/app.py

Live demo: change the domain, app and date filters → KPIs and charts update; pick a topic → its terms and real reviews; switch the app on Forecasting; paste a review into Check a Review.

Business scenario: "Paytm's complaints jumped in September: why, and what do we do?" Forecasting (Paytm) shows alerts after 3 Sep and a red release marker → Update impact confirms 10.84.5 → Recommendations: High priority, investigate the release.



Completion and individual contribution

Team member	Completed responsibility	Evidence / project-board task
Aditya Monish Kumar K (CB.SC.U4CSE23103)	R1: scraper, collection, cleaning, integrity tests. R2: time-series and text preparation, dashboard base and deployment; validation, report, deck	#53, #54, #56–#58 (PR #52, #99); #131–#133 (PR #147–#149)
Regella Krishna Saketh (CB.SC.U4CSE23649)	R1: EDA and bias audit (with Akshay). R2: NMF topic models, sentiment per topic, coherence, keyword-tag comparison	#104–#106, #118–#120 (PR #121); #134, #136 (PR #150, #151)
Akshay KS (CB.SC.U4CSE23104)	R1: EDA and bias audit (with Saketh). R2: ARIMA/SARIMA forecasting with rolling backtest; text-mining page	#104–#106, #118–#120 (PR #121); #135, #139 (PR #152, #153)
Harshini Vennela (CB.SC.U4CSE23455)	R1: features and classifiers (with Kanishka). R2: forecast evaluation and alerts, forecast write-up; check-a-review page	#107–#110 (PR #122–#125); #137, #142, #146 (PR #154–#156)
Kanishka D (CB.SC.U4CSE23155)	R1: features and classifiers (with Harshini). R2: update impact, combined insights; forecasting, update-impact and recommendations pages; final notebook	#107–#110 (PR #122–#125); #138, #140, #141, #143 (PR #157, #158)

Project board: github.com/users/Adithya-Monish-Kumar-K/projects/4 (29 of 29 tasks Completed) **Repository / deliverables:** github.com/Adithya-Monish-Kumar-K/ba-capstone-app-reviews

Conclusion and limitations

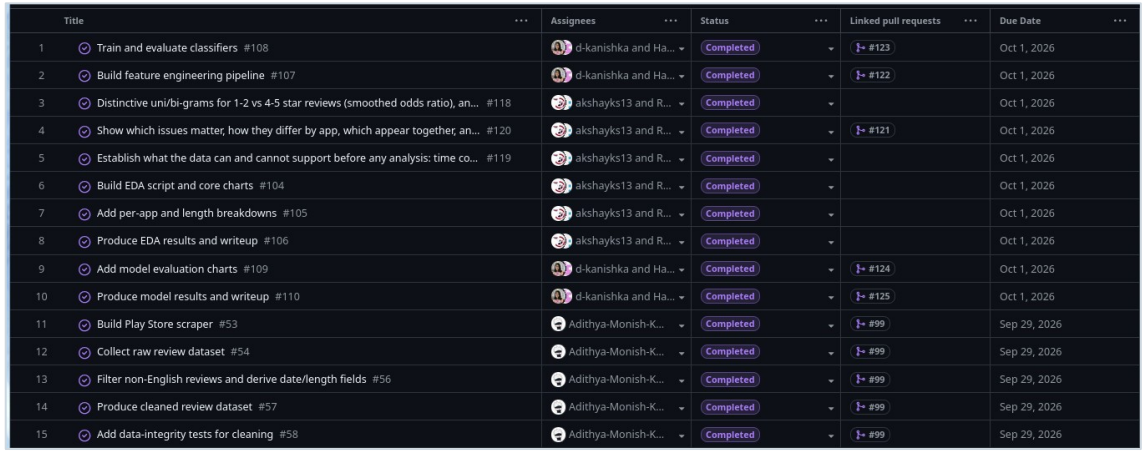
Main outcome: a complete pipeline from 1.14 million self-collected reviews to a problem-review classifier (PR-AUC 0.87), 17 complaint topics, ARIMA forecasts that beat the baseline for 10 of 11 apps, release verdicts and a priority table, live in a dashboard.

Business value: the biggest fixes are refunds, returns and failed transfers; Paytm 10.84.5 explains the September rise; Swiggy, Flipkart, Amazon and Paytm come first; next week's complaint volume is forecast with an interval.

Limitations: 173 days, so no yearly season; a 1–2★ review is the user's judgement, not a confirmed defect; VADER and keyword tags miss context; the feed gap and users choosing when to update limit the release tests.

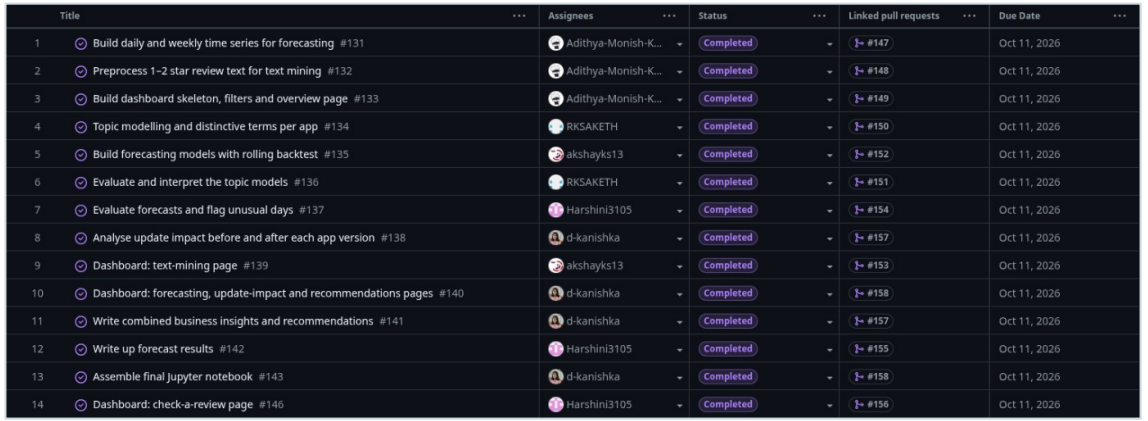
Future scope: daily automatic collection and alerts; Hindi and Hinglish text models; store release notes to confirm release dates; the iOS App Store.

Project-board task history (Reviews 1 and 2)



	Title	Assignees	Status	Linked pull requests	Due Date
1	Train and evaluate classifiers #108	d-kanishka and Ha...	Completed	#123	Oct 1, 2026
2	Build feature engineering pipeline #107	d-kanishka and Ha...	Completed	#122	Oct 1, 2026
3	Distinctive uni/bi-grams for 1-2 vs 4-5 star reviews (smoothed odds ratio), an... #118	akshayks13 and R...	Completed		Oct 1, 2026
4	Show which issues matter, how they differ by app, which appear together, an... #120	akshayks13 and R...	Completed	#121	Oct 1, 2026
5	Establish what the data can and cannot support before any analysis: time co... #119	akshayks13 and R...	Completed		Oct 1, 2026
6	Build EDA script and core charts #104	akshayks13 and R...	Completed		Oct 1, 2026
7	Add per-app and length breakdowns #105	akshayks13 and R...	Completed		Oct 1, 2026
8	Produce EDA results and writeup #106	akshayks13 and R...	Completed		Oct 1, 2026
9	Add model evaluation charts #109	d-kanishka and Ha...	Completed	#124	Oct 1, 2026
10	Produce model results and writeup #110	d-kanishka and Ha...	Completed	#125	Oct 1, 2026
11	Build Play Store scraper #53	Adithya-Monish-K...	Completed	#99	Sep 29, 2026
12	Collect raw review dataset #54	Adithya-Monish-K...	Completed	#99	Sep 29, 2026
13	Filter non-English reviews and derive date/length fields #56	Adithya-Monish-K...	Completed	#99	Sep 29, 2026
14	Produce cleaned review dataset #57	Adithya-Monish-K...	Completed	#99	Sep 29, 2026
15	Add data-integrity tests for cleaning #58	Adithya-Monish-K...	Completed	#99	Sep 29, 2026

Review 1: 15 tasks (data, EDA, models), due 29 Sep – 1 Oct, all Completed

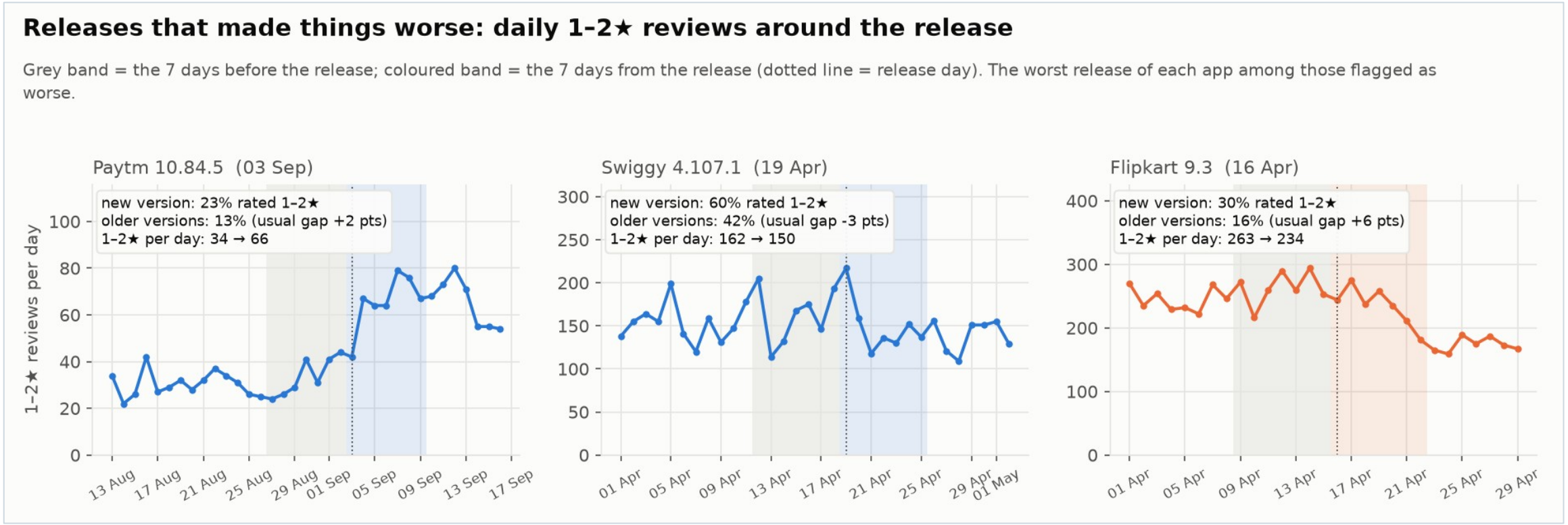


	Title	Assignees	Status	Linked pull requests	Due Date
1	Build daily and weekly time series for forecasting #131	Adithya-Monish-K...	Completed	#147	Oct 11, 2026
2	Preprocess 1–2 star review text for text mining #132	Adithya-Monish-K...	Completed	#148	Oct 11, 2026
3	Build dashboard skeleton, filters and overview page #133	Adithya-Monish-K...	Completed	#149	Oct 11, 2026
4	Topic modelling and distinctive terms per app #134	RKSAKETH	Completed	#150	Oct 11, 2026
5	Build forecasting models with rolling backtest #135	akshayks13	Completed	#152	Oct 11, 2026
6	Evaluate and interpret the topic models #136	RKSAKETH	Completed	#151	Oct 11, 2026
7	Evaluate forecasts and flag unusual days #137	Harshini3105	Completed	#154	Oct 11, 2026
8	Analyse update impact before and after each app version #138	d-kanishka	Completed	#157	Oct 11, 2026
9	Dashboard: text-mining page #139	akshayks13	Completed	#153	Oct 11, 2026
10	Dashboard: forecasting, update-impact and recommendations pages #140	d-kanishka	Completed	#158	Oct 11, 2026
11	Write combined business insights and recommendations #141	d-kanishka	Completed	#157	Oct 11, 2026
12	Write up forecast results #142	Harshini3105	Completed	#155	Oct 11, 2026
13	Assemble final Jupyter notebook #143	d-kanishka	Completed	#158	Oct 11, 2026
14	Dashboard: check-a-review page #146	Harshini3105	Completed	#156	Oct 11, 2026

Review 2: 14 tasks (both methods, dashboard, insights), due 11 Oct, all Completed

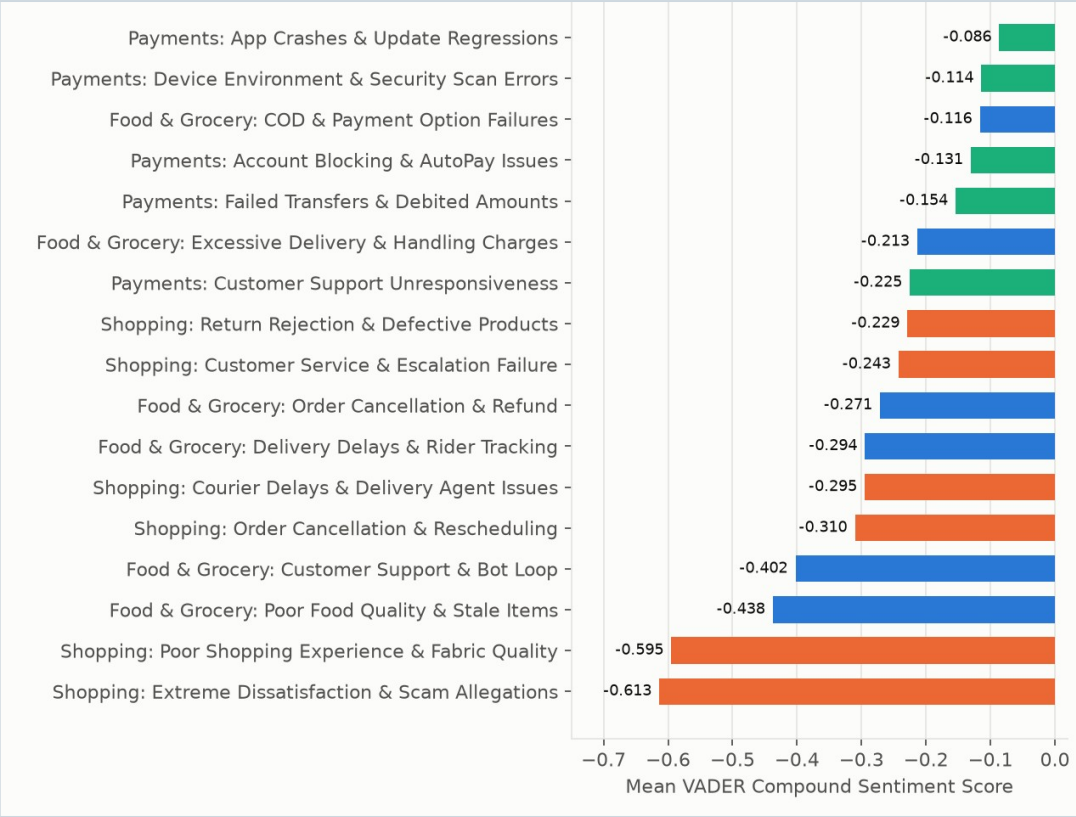
Activity history: Review 1 merged 28–30 Sep. Review 2 as a relay, each stage started on the merged work of the previous one: Stage 1 and 2 on 8 Oct, Stage 3 on 9 Oct, Stage 4 on 10 Oct, Stage 5 on 11 Oct. Every card moved Backlog → To Do → In Progress → Review/Testing → Completed; each is linked to its merged pull request.

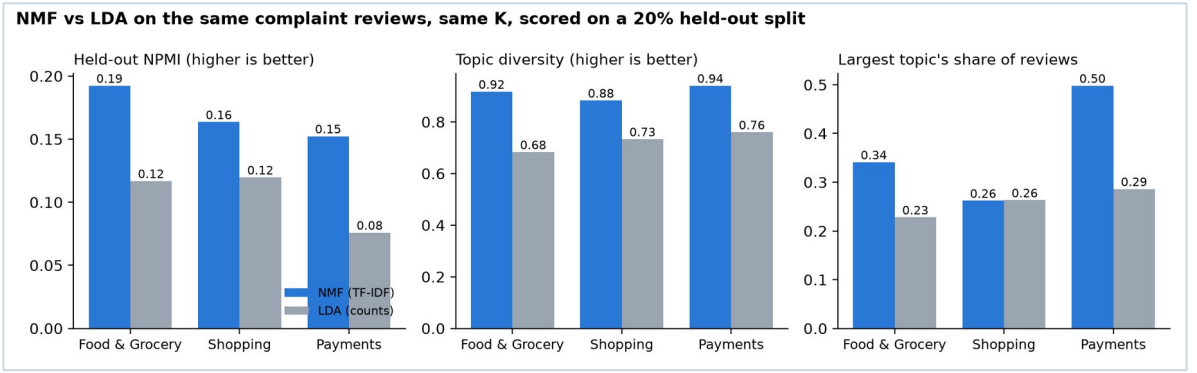
Update impact: releases that made things worse



224 releases, 205 testable: 196 no clear change, 5 better, 4 worse. Paytm 10.84.5 (3 Sep) is the only one confirmed at app level: 23% vs 13% 1–2★ on the new version, 34 → 66 1–2★ reviews a day. Swiggy 4.107.1 / 4.107.2 and Flipkart 9.3 overlap the feed gap, so their shares are inflated.

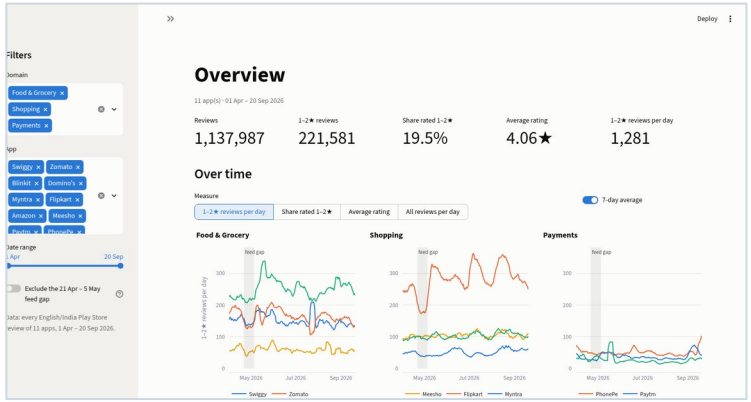
Sentiment by complaint topic, and NMF vs LDA



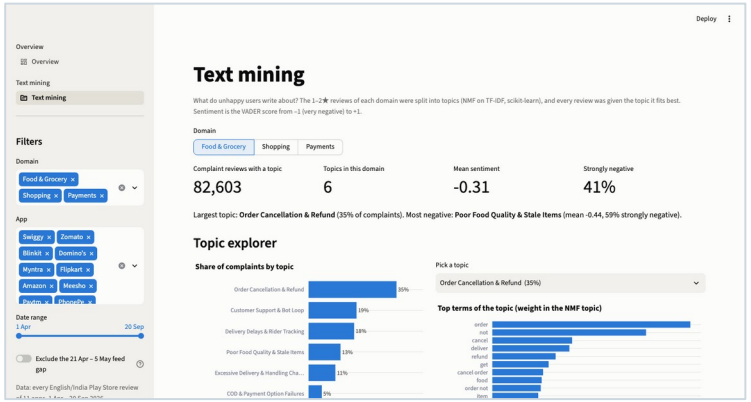


Angriest topics: scam allegations (−0.61) and poor product quality (−0.60) in shopping; stale food (−0.44) angers more than late food (−0.29).
Calm but serious: payment app crashes (−0.09) are described factually, so tone is not severity.
NMF vs LDA: same data and K, 20% held out; NMF more coherent and more distinct in all 3 domains (scripts/23_topic_model_comparison.py).

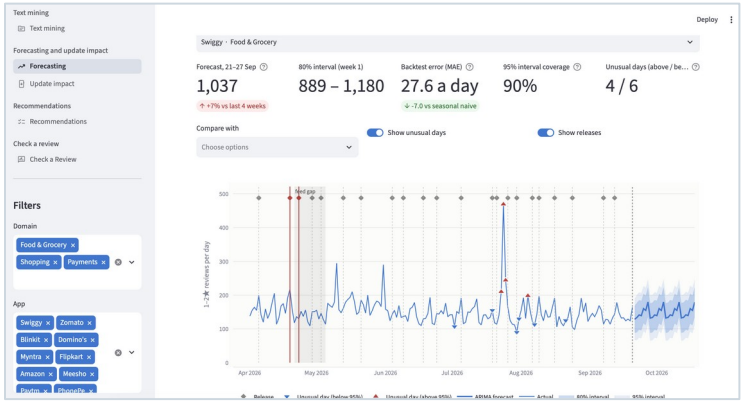
Dashboard pages



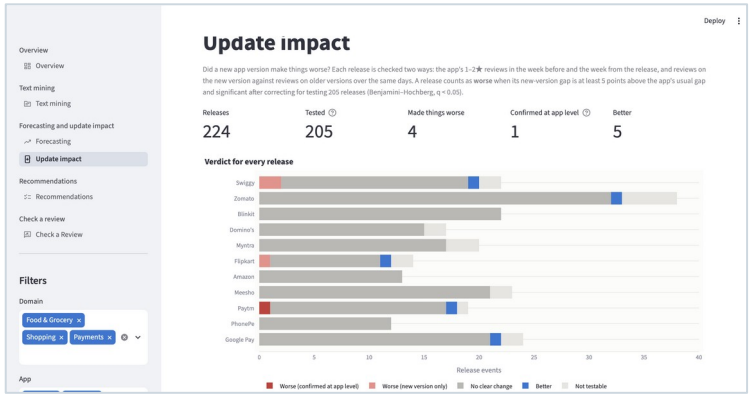
Overview



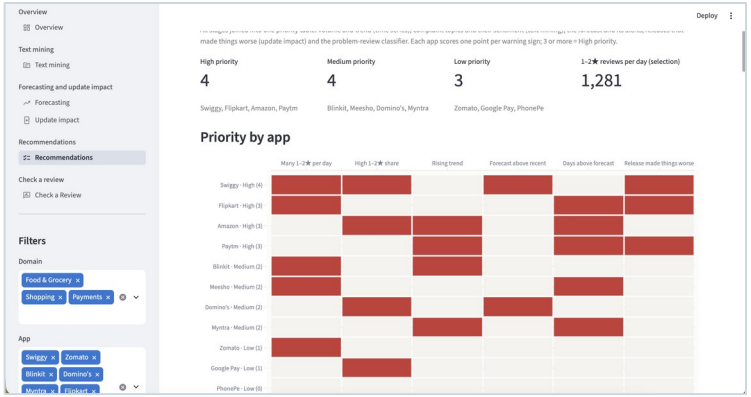
Text mining



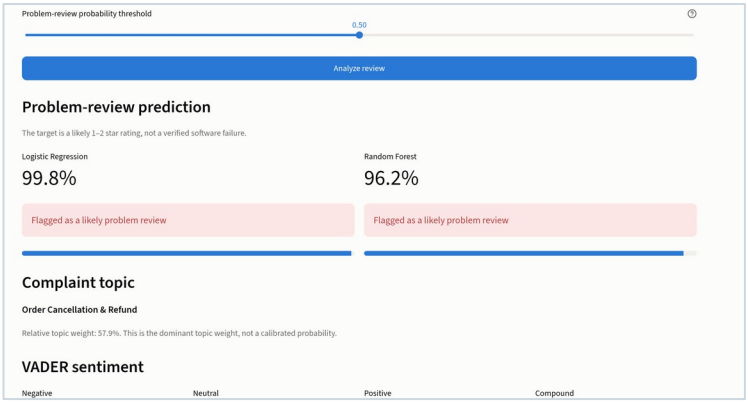
Forecasting



Update impact



Recommendations



Check a review